Mathematical Foundations of Political Methodology
Lecture 3

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Sequences and Limits

- We characterize the behavior of smooth functions with limits.
- To understand a limit, we need to have a unit of account, which is a sequence.

- Sequence
- Cauchy Sequence
- Limits of Functions
- Continuity

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Sequence

Sequence can be irregular, an example of such sequence is:

$$a_1 = 1$$

$$a_2 = 8$$

$$a_3 = 6$$

$$a_4 = ...$$

Sequence can be regular, an example of such sequence is:

$$x_1 = 2$$

$$x_2 = 4$$

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- The sequence is written $\{u_n\}$
- For example:

$$u_n = \frac{3n^2}{n^3 + 9}$$

gives the sequence

$$\frac{3}{10}, \frac{12}{17}, \frac{48}{73}, \ldots$$

Definition

A sequence of a set X is a function, u, mapping $\mathbb{N}$ to X

- Ideal classes of sequences:
    - Arithmetic sequences
    - Geometric sequences
- Defining characteristics of sequences
    - Recurrent behavior: oscillation, convergence, divergence
    - Specific kinds of convergence.

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- Is this sequence arithmetic? (If so, what are a, d?)

$$u_n = 14 - 5n$$

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- What are a, x for this geometric sequence?

$$u_n = 5^{-n}$$

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- How do we measure the behavior of sequences that change over time?
- Many sequences behave similarly as n gets large.
- One way to characterize these sequences is to identify where they are converging to, a limit.
- However, we need an objective criteria for determining whether two sequences are going to the same place.

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- The sample average with increasing sample size will converge (in probability) to the true mean.

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Also written as

$$\lim u_n = z$$

$$\lim_{n \to \infty} u_n = z$$

$$u_n \to z \text{ as } n \to \infty$$

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  - If a sequence converges, it converges to one number, say z.
  - We check by picking an $\epsilon > 0$, which is an arbitrary real-valued number as the maximum deviation from the proposed limit z.
  - Then we find the N, which will depend upon ϵ, where the sequence is within the ϵ region of z.

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For each ϵ, then, any $n > N_\epsilon > \frac{1}{\epsilon}$ will suffice. $\square$

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We note then, that any $n > N_\epsilon > \frac{1}{\epsilon^2}$ will work. $\square$

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$$|\frac{3}{n^5}| < \epsilon$$

Solve for N_ϵ:

$$|\frac{3}{\epsilon}| < N_\epsilon^5$$

$$\left(\frac{3}{\epsilon}\right)^{\frac{1}{5}} < N_\epsilon$$

For every $n > N$, the inequality $n > N_\epsilon > \frac{3}{\epsilon}^{\frac{1}{5}}$, so $n > \left(\frac{3}{\epsilon}\right)^{\frac{1}{5}}$. So all integers $n > N_\epsilon > \left(\frac{3}{\epsilon}\right)^{\frac{1}{5}}$ make $\frac{3}{n^5} < \epsilon$.

END VIDEO

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Cauchy, (pronounced KohShee) Sequence: Motivation

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A popular test for the existence of a limit is to determine if that sequence is Cauchy.

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- A sequence that tends to infinity (or negative infinity) is divergent, but a sequence doesn't have to tend to infinity to be divergent. (What's an example of a divergent sequence that doesn't tend to infinity?)

Informal Definition of Cauchy Sequence

A sequence $\{x_n\}$ is Cauchy if it eventually stops bouncing around more than ϵ:

- Find two points in the sequence x_m and x_n.
- Take their difference. Is it getting smaller as m and n get big?

Formal Definition of Cauchy Sequence

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Definition

$\{x_n\}$ *is Cauchy if* $\forall \epsilon > 0, \ \exists N \in \mathbb{N}:$
$|x_m - x_n| < \epsilon$ *when* $m, n > N$

Theorem

In $\mathbb{R}$, $\{x_n\}$ *is convergent if and only if it's Cauchy.*

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- Thus, for $\epsilon = 1$ there is no N such that $|x_n - x_m| < \epsilon$ whenever $n, m > N$.
- Therefore $\{x_n\}$ is not Cauchy, and does not tend to a limit.

Working with Limits

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- Fact 2: For a real number b, $\lim_{n\to\infty} n^b = 0$ if and only if $b < 0$.

Working with Limits: Evaluate Dominating Term

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Don't do whole calculation!

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- Thus, series is a special type of sequence, it sums the terms in the sequence.

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Suppose $j = 0$, $n = 10$ and $u_r = 2 * r$

$$s_n = \sum_{r=0}^{10} u_r = 0 + 2 + 4 + 6 + 8 + 10 + 12 + 14 + 16 + 18 + 20$$

$$= \sum_{r=0}^{3} u_r + \sum_{r=4}^{7} u_r + \sum_{r=8}^{10} u_r$$

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The first n terms of the natural numbers sum to

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$$u_n = an + n\left(\frac{(n-1)d}{2}\right)$$

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Then $s_n = a + t$ and $xs_n = t + ax^n$, we can eliminate t.

$$s_n - xs_n \quad = \quad a - ax^n$$

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Then $s_n = a + t$ and $xs_n = t + ax^n$, we can eliminate t.

$$\begin{aligned} s_n - xs_n &= a - ax^n \\ s_n(1 - x) &= a(1 - x^n) \end{aligned}$$

Summing Geometric Sequences

These same tricks can be extended to the geometric sequence:

$$s_n = a + ax + ax^2 + ... + ax^{n-1}$$

Our strategy is to construct a forward and backward shifted sequence and then take differences to eliminate the middle.

Multiplying by x we get our original sequence shifted forward.

$$xs_n = ax + ax^2 + ... + ax^n$$

We then define a sequence for all but the first term.

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Then $s_n = a + t$ and $xs_n = t + ax^n$, we can eliminate t.

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Sum them together and you will have your total earnings:

$1000*(1+0.05)^n + 1000*(1+0.05)^{n-1} + ... + 1000*(1+0.05)^1 = 13,206.79$

Utility Application of Geometric Series

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- We can use the properties of series to describe the long run (infinite) value of a choice.

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For a geometric series,

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Applying this result to the long run value of a:

$$U(a) = a + a\delta + a\delta^2 + ... + a\delta^{n-1} = \frac{a}{1-\delta}$$

$$\sum_{t=0}^{\infty} u(x)\delta^t = \frac{u(x)}{1-\delta}$$

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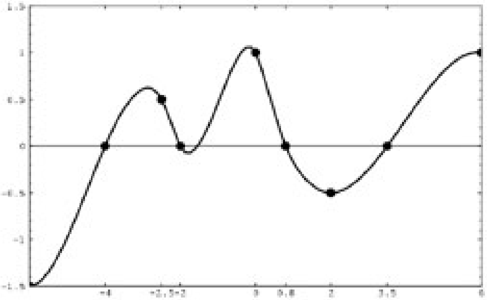
Therefore, a geometric series diverges whenever $|x| \geq 1$

END VIDEO

- Sequence
- Cauchy Sequence
- Limits of Functions
- Continuity

Limits of Functions: Motivation



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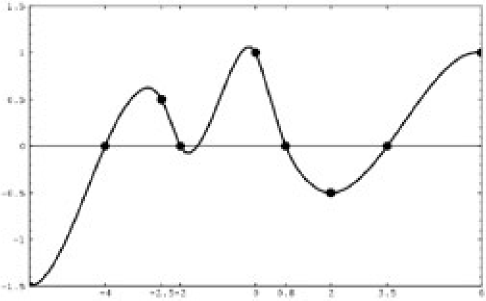

We care about the limits of function because we care about how functions behave at a point.
It turns out that function can often only be described in the limit (recall probability example).

Limits of Functions

Definition

Suppose $f : \mathbb{R} \to \mathbb{R}$. *We say that* f *has a limit* L *at* x_0 *if, for each* $\epsilon > 0$, *there is a* $\delta > 0$ *such that* $|x - x_0| < \delta$ *implies that* $|f(x) - L| < \epsilon$.

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- δ defines an area around x_0 where $f(x)$ is going to be within our error rate

Limits of Functions: Example

Theorem

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Proof step 1, simplify

For all $x \neq 1$,

$$\begin{aligned} f(x) &= \frac{x^2-1}{x-1} \\ &= \frac{(x+1)(x-1)}{x-1} \\ &= x+1 \end{aligned}$$

Limits of Functions: Example Continued

Show $f(x) = \frac{x^2-1}{x-1} = x + 1$ has a limit of 2 at $x_0 = 1$.

Proof step 2:

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$$f(x) = \begin{cases} +1 & \text{if } x < 0 \\ -1 & \text{if } x \geq 0 \end{cases}$$

Algebra of Limits

Theorem

Suppose $f : \mathbb{R} \to \mathbb{R}$ *and* $g : \mathbb{R} \to \mathbb{R}$ *with limits* A *and* B *at* x_0*. Then,*

$$i.)\ \lim_{x \to x_0} (f(x) + g(x)) = \lim_{x \to x_0} f(x) + \lim_{x \to x_0} g(x) = A + B$$

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Suppose $g(x) \neq 0$ *for all* $x \in \mathbb{R}$ *and* $B \neq 0$ *then* $\frac{f(x)}{g(x)}$ *has a limit at* x_0 *and*

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \frac{\lim_{x \to x_0} f(x)}{\lim_{x \to x_0} g(x)} = \frac{A}{B}$$

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- It has special properties when we get to derivatives.

END VIDEO

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Definition

Suppose $f : \mathbb{R} \to \mathbb{R}$ and consider $x_0 \in \mathbb{R}$. We will say f is continuous at x_0 if for each $\epsilon > 0$ there is a $\delta > 0$ such that if,

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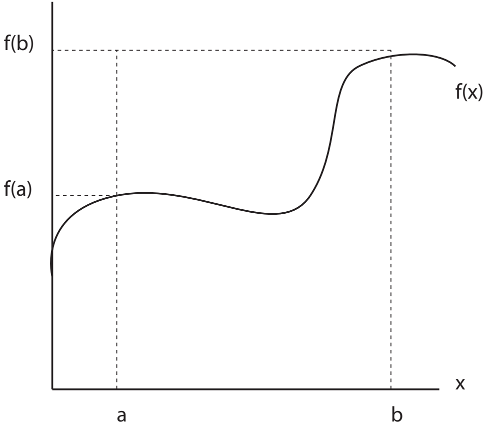
- Continuity is more restrictive than limit.

The intermediate value theorem

Consider an interval $[a, \ b]$ in the real numbers.
If the function $f(x)$ is continuous for all $a \leq x \leq b$ then f takes every value between $f(a)$ and $f(b)$.



Formal Statement of the intermediate value theorem

Let $f(x)$ be a function which is continuous on the interval $[a, b]$, let u be a real number between $f(a)$ and $f(b)$, there is at least one c such that $a \leq c \leq b$ such that $u = f(c)$

Applications of intermediate value theorem: mashed potato

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- A plate of mashed potato can be evenly divided by a single straight vertical knife cut.
  Proof: In position K_1 less than half the potato is at the left of the knife, in position K2 more than half is at the left. Hence (by the intermediate value theorem) there is an intermediate position where exactly half is at one side.

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4) You passed through an intermediate position where both beans and potatoes were divided fairly.

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- Pick a small area around $f(x)$, use properties of continuity of f and the real numbers, find that the only value that can satisfy these conditions is $f(c) = u$

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- Example: If $S = \{x : x > 3\}$, $\nexists \sup S$.

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S is nonempty, as $a \in S$.

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Consider $S = \{x \in [a, \ b] : \ f(z) \leq u \text{ for all } z \in [a, x]\}$

S is nonempty, as $a \in S$.

S has an upper bound, b.

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S has an upper bound, b.

Therefore S has a least upper bound, we claim it is c and $f(c) = u$

Proof of intermediate value theorem step 2

Fix some $\epsilon > 0$

- By continuity, there is a $\delta > 0$ such that $|x - c| < \delta$ implies $|f(x) - f(c)| < \epsilon$.

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- Unpacking the absolute value bars:

$$\begin{aligned} |f(x) - f(c)| &< \epsilon \text{ if } |x - c| < \delta \\ -\epsilon < f(x) - f(c) &< \epsilon \text{ if } -\delta < x - c < \delta \\ f(x) - \epsilon < f(c) &< f(x) + \epsilon \text{ if } x \in (c - \delta,\ c + \delta) \end{aligned}$$

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$$f(x) - \epsilon < f(c) < f(x) + \epsilon \text{ if } x \in (c - \delta,\ c + \delta)$$

- Because c is the supremum, there is some $d_1 \in (c - \delta, c]$ that is in S, so

$$f(c) < f(d_1) + \epsilon$$
$$f(d_1) + \epsilon < u + \epsilon$$

- There is some $d_2 \in (c, c + \delta)$ where $d_2 \notin S$,

$$f(d_2) - \epsilon < f(c)$$

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$$f(c) < f(d_1) + \epsilon < u + \epsilon$$
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For all ϵ.

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For all ϵ.
So $f(c) = u$